

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

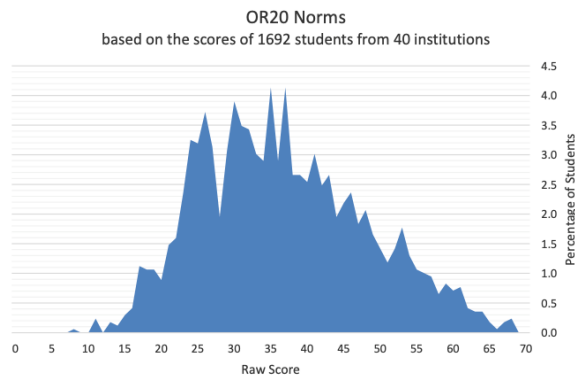
CIM code (e.g., ME-CM)	BCHM
Program Name	B.S. in Biochemistry
Assessment Year	2024-2025
Program Assessment Coordinator	David Minh and Andrew Howard
Year of Current Assessment Plan	2023
No Assessment This Year per Plan	☐
LO(s) Assessed	#4 Organic Chemistry
Presented at program authority meeting date: <i>Please attach minutes</i>	Scheduled for October 31, 2025

Methodology

For each LO and artifact assessed, please describe methodology used, and attach (as supplemental files during report submission) or insert here the rubric used. At minimum this should contain:

- a description of how the artifact is concordant with the LO
- a rubric describing the method used to place individual student in at least 3 assessment categories; {**D** Does not meet expectations, **M** Meets expectations **M**, **P** Proficient}

The artifact was the final exam grade in the second semester of an organic chemistry course, which is clearly concordant with the LO in organic chemistry. The exam was from the American Chemical Society. The national mean of scores was 36.7 and standard deviation was 11.3. To meet expectations, students should score higher than one standard deviation below the mean (25.4). To be proficient, students should score higher than one standard deviation above the mean (48.0).



There were two biochemistry majors in the course, so results are limited to these students.

Results

Artifact samples

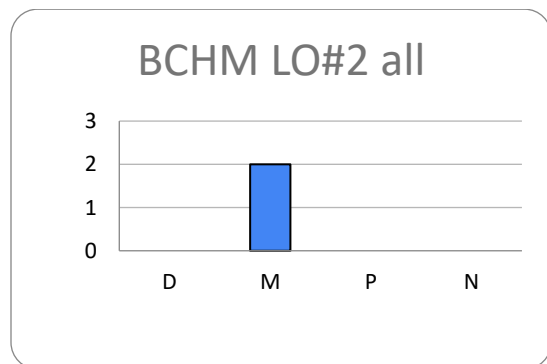
Attach artifact samples. You can paste them in here if feasible, attach them to the end of this report, or submit them as supplemental files when submitting your reports. For data sets with ≤ 10 students, provide 1 in each category, D,M,P, (unless the category is empty, if so state that). For programs with ≤ 20 , provide 2. For other programs provide 3. Please redact any personally identifiable information, PII (which includes names and A#s).

Grades for the two students were **41/70** and **29/70**. The ACS does not release question-by-question results to the member institutions, so we cannot provide artifacts beyond their composite scores.

Data table

For each LO and provide a table pasted in from the downloadable [provided template](#), placing each student into the assessment categories (minimally {D, M, P}). You should collect and analyze data linked to student A#s locally, but please redact any personally identifiable information, PII (including names or A#s) in tables presented in this report. Provide a summary graph from the template presenting this distribution. You should also present additional analysis as deemed appropriate by the results seen, and the needs of the program.

student	Major	D/M/P
student 1	BCHM	M
student 2	BCHM	M



Description of findings

Summarize the discussion that occurred within the program authority when this was presented. Provide a short narrative interpreting the findings. Are there any areas of concern?

The discussion has not occurred yet. However, a concern was raised over email about the lack of proficient students (as both students only met expectations) and the small program size that limited the statistical value of the data. At the next meeting of the Biochemistry Program Steering Committee, we plan to raise the issue of accumulating data over multiple years rather than selecting out a single year. Having only two Biochemistry students enrolled in Organic Chemistry is somewhat anomalous, but the number per year is unlikely to increase dramatically.

Continuous Improvement, CI, plans

The purpose of assessment is to improve the quality of our educational programs. The most important part of the assessment cycle is how we use the data to improve our programs. Examine the data provided above describing student achievement, consider the root cause of any deficits, and if areas for possible improvement emerge, describe plans to achieve this improvement.

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

☒ no previous CI plans available

No CI plans relating to this LO have been previously recommended or implemented.

This cycle's CI plan

Describe any CI plans developed from this cycle's data, with

- Specific curricular or instructional changes in response to this assessment data,
- A timeline for implementation,
- As well a description of who will participate in and manage these plans

☒ After consideration and presentation to the program authority, no CI plans are recommended.

Based on the data that we have obtained, there is no need to modify the program to improve the proficiency of biochemistry majors in organic chemistry. However, this result has limited statistical value and is not consistent with the experience of faculty teaching a more advanced course, BIO 401. We recommend that the curriculum map within the assessment plan be modified to evaluate this LO based on artifacts from BIO 401. We also recommend that the assessment plan be modified to collect data over multiple years.